

Name: _____

Class: _____

ACTIVITY SHEET

4.4 Reactivity of metals

The table below summarises observations of the reactivity of different metal elements.

Element	K	Na	Li	Ca	Mg	Al	Zn	Fe	Pb	Cu	Ag	Au
In air at 20°C	rapidly tarnishes	rapidly tarnishes	rapidly tarnishes	no observable reaction								
Heated in oxygen	burns vigorously	burns vigorously	burns strongly	burns quite fast	burns vigorously	goes white	goes yellow	moderate reaction	slow reaction	surface blackens	no reaction	
In water at 20°C	reacts very violently	reacts violently	fast reaction	reacts readily	reacts very slowly	no observable reaction						
In heated steam	explosive	explosive	very violent	violent	burns rapidly	little reaction	moderate reaction	reacts slowly	no observable reaction			
In dilute HCl, and H ₂ SO ₄	explosive reaction	explosive reaction	violent reaction	reactive with HCl	reactive with both	slow reaction	quite reactive	reacts slowly	no observable reaction			

- What chemical would be causing K, Na and Li to tarnish when exposed to the air?
- State two reasons why most metals react more vigorously when heated in oxygen gas than when exposed to the air.
- Write a general word equation for the reaction of reactive metals with oxygen gas.
- Write balanced equations for the reactions of the following metals with oxygen.
 - Lithium
 - Calcium
 - Aluminium

- 5 Write a general word equation for the reaction of reactive metals with water.
- 6 Explain why reactive metals react more vigorously with heated steam than with water at 20°C.
- 7 Write balanced equations for the reactions of the following metals with water at room temperature.
- a Sodium
 - b Magnesium
 - c Zinc
- 8 Write a general word equation for the reaction of reactive metals with acids.
- 9 Suggest a reason why the metals are reacted with dilute rather than concentrated acids.
- 10 Write balanced equations for the reactions of the following metals with dilute hydrochloric acid.
- a Lithium
 - b Copper
 - c Magnesium
- 11 Write balanced equations for the reactions of the following metals with dilute sulfuric acid.
- a Sodium
 - b Calcium
 - c Aluminium

12 Write down the reactivity series for the metals listed in the table.

13 Which metal does not appear to react as strongly as it ought to, according to the reactivity series?

14 Suggest why that metal does not react as predicted.